Look at a diagram of the heart and use it to help you answer the following questions.

1) State the full names of the heart's chambers.

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2) Choose the correct endings for each sentence. Underline your choice in each bracket.

The top two chambers have these similarities:

- They are both (ventricles/atria).
- They both (receive blood into the heart / send blood out of the heart).
- They both send blood to (the body / the lungs / a ventricle).

3) Describe the similarities between the heart's two bottom chambers.

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4) Suggest what is unusual about the way the chambers are named.

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* Suggest a reason for this.

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5) Describe the differences in the structure and function of the two ventricles.

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6) Select the word that describes the blood in the left side of the heart. Underline your choice.

deoxygenated	red	blue	oxygenated

7) Explain why the blood on the right side of the heart needs to go to the lungs.

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8) Suggest which structures in the heart prevent the backflow of blood.

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* Why do you think this is important?

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9) * Describe the journey of a red blood cell from leaving the pulmonary artery until reaching the aorta.

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10)* Summarise the structure and function of the heart in no more than 100 words.

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Teaching notes

These questions were designed to be used with the jigsaw ‘The heart’ (19999). However, they could be used with other heart diagrams. Some modification of the questions may be necessary in this case.

Questions which are designed to provide more challenge are marked with an asterisk.

Suggested answers:

1) Right atrium, left atrium, right ventricle, right atrium

2) For each bullet point, the following should be selected:

- atria
- receive blood into the heart
- a ventricle

Note: the word ‘atria’ is not used on the diagram, but this choice can be deduced and is a literacy learning point.

3) Similarities:

- They are both ventricles
- They both receive blood from atria
- They both send blood out of the heart.

4) The chambers on the left of the diagram are named ‘right, the chambers on the right of the diagram are named ‘left’.

   *This is because if the heart was in your body, then the sides would be reversed – they would be on the correct sides.

Note: pupils usually understand the logic behind this better if they pretend to transplant the heart diagram into themselves.

5) Differences:

   The left ventricle has a thicker wall than the right ventricle. The left ventricle sends blood out of the heart through the aorta rather than the pulmonary artery. The left ventricle is pumping out (oxygenated) blood to the body, whereas the right ventricle is pumping out (deoxygenated) blood to the lungs.

6) Oxygenated

7) The blood is pumped to the lungs because it is deoxygenated, meaning it is low in oxygen. In the lungs, the haemoglobin in the red blood cells will be able to pick up oxygen.

Note: discourage pupils saying that the blood has no oxygen.

8) Valves.

*This is important to make the heart as efficient as possible e.g. if blood could flow backwards, less blood may be in the ventricles to be pumped out and extra work would be needed to get the blood back into the atria. Blood might back up in the atria. Pupils sometimes find it useful for the valves to be equated to traffic police controlling the flow of cars.

9) *It leaves the pulmonary artery (deoxygenated), goes to the capillaries in the lungs (and becomes oxygenated). It returns to heart through the pulmonary vein and goes into the left atrium. It goes through a valve and into the left ventricle. It is pumped out of heart through the aorta.

10) The heart has two sides with two chambers on each. The top chambers are called atria. They receive blood. Blood then flows through valves into the ventricles (bottom chambers). Valves stop the blood flowing backwards. The right side receives deoxygenated blood from the body through the vena cava. This blood exits through the pulmonary artery to the lungs.
The left side receives oxygenated blood from the lungs through the pulmonary vein and exits through the aorta to the body. The left ventricle has a thicker wall to pump blood to the rest of the body.

Note: It is quite challenging to include everything in 100 words. You may wish to suggest bullet points or extend the word limit!